Master's Thesis Progress Documentation

Phase 1: Research Foundation, Topic Clarification, and Initial Planning

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1. Thesis Title

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

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2. Purpose of This Document

This document summarizes the work completed during the first phase of the Master's thesis. The main focus of this phase was to understand the research topic clearly, define the problem area, identify the technical and visualization components, and prepare the foundation for the next implementation steps.

This document is part of the project documentation and will be maintained in the GitHub repository to track the development of the thesis work in a structured and transparent way.

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3. Current Stage of the Thesis

The thesis is currently in the initial planning and research foundation stage.

At this stage, the main goal was not yet to build the final system, but to define the direction clearly and prepare the required background before moving into dataset selection, model development, explainability implementation, and Power BI dashboard design.

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4. Work Completed in Phase 1

During the first phase, the following work was completed:

4.1 Topic Finalization

The thesis topic was finalized as:

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

The topic was selected because it combines three important areas:

- Machine learning prediction
- Explainable Artificial Intelligence

- Interactive visual analytics

The focus is to create a system where a predictive model generates outputs, explainability methods explain those outputs, and Power BI is used to present the results in a clear and interactive visual format.

4.2 Understanding the Research Problem

A key part of this phase was understanding the main problem behind the thesis.

In many business environments, machine learning models are used to make predictions such as customer churn, credit risk, or demand forecasting. However, these models usually provide outputs such as probabilities or classification labels.

For example, a model may predict:

Customer churn probability: 82%

Although this value is useful, it does not fully explain why the customer is likely to churn. A business user may still need to understand:

- Which factors influenced the prediction
- Whether the prediction is reliable
- Which variables increased or decreased the risk
- How the result changes under different conditions

This creates the need for explainable and visually interpretable AI systems.

4.3 Clarification of the System Concept

The planned system was defined as a pipeline with the following structure:

Dataset → Machine Learning Model → Explainability Method → Structured Explanation Outputs → Power BI Dashboard → User Interpretation

The machine learning model will be developed in Python. The explainability outputs, such as feature contribution values, will be exported and then integrated into Power BI.

Power BI will act as the main visual analytics interface where users can explore predictions and explanations interactively.

4.4 Clarification of Power BI Role

Since the thesis has a strong visualization component, Power BI was selected as the main dashboard and interaction tool.

The purpose of Power BI in this thesis is not to automatically generate visualizations using AI. Instead, Power BI will be used to intentionally design an interactive visual interface that helps users understand model predictions.

The dashboard will include elements such as:

- Prediction scores
- Feature contribution charts
- Positive and negative influence indicators
- Segment-level comparisons
- Filters and drill-downs
- Possible what-if analysis

This makes Power BI the human-facing layer of the system.

4.5 Initial Evaluation Design

The evaluation approach was also clarified during this phase.

The goal of the evaluation is to compare two versions of the system:

Version A: Static Baseline
A non-interactive version showing predictions and explanation charts.

Version B: Interactive Power BI Dashboard
An interactive version allowing users to filter, compare, and explore prediction explanations.

The evaluation will measure:

- Interpretation accuracy
- Time taken to complete tasks
- User confidence
- Trust in the system
- Decision quality

This comparison will help determine whether interactive visual analytics improves understanding and decision-making compared to static explanations.

5. Planned Tools and Technologies

The following tools and technologies are currently planned:

Python

Python will be used for:

- Data preprocessing
- Machine learning model development
- Model evaluation
- Explainability calculation

Machine Learning Libraries

Possible libraries include:

- scikit-learn
- pandas
- numpy
- SHAP

Power BI

Power BI will be used for:

- Dashboard development
- Interactive visual analytics
- Feature contribution visualization
- User-facing decision-support interface

GitHub

GitHub will be used for:

- Tracking progress
- Version control
- Project documentation
- Storing code files
- Maintaining PDF documentation

6. Initial Literature Areas Identified

The following literature areas have been identified for review:

- Explainable Artificial Intelligence
- SHAP and feature attribution methods
- Machine learning interpretability
- Visual analytics
- Business intelligence dashboards
- Human-AI interaction
- Trust and usability in AI systems

These areas will support the theoretical foundation of the thesis.

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7. Next Steps

The next phase of the thesis will focus on moving from planning to implementation.

The planned next steps are:

1. Select and finalize a suitable business dataset.
2. Perform exploratory data analysis.
3. Clean and preprocess the dataset.
4. Train an initial machine learning model.
5. Evaluate model performance.
6. Apply explainability methods such as SHAP.
7. Export prediction and explanation outputs.
8. Start designing the Power BI dashboard structure.

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8. Summary

The first phase of the thesis focused on topic clarification, research understanding, system planning, and evaluation design.

The thesis direction is now clearly defined. The project will combine machine learning, explainable AI, and Power BI-based visual analytics to improve the interpretability of AI predictions in business decision-making contexts.